



Course: C++ Programming

From Fundamentals to Object-Oriented Programming

Duration

- 60 Hours Training
- 90 Hours with Lab Practice
- 120 Hours Advanced Program

Prerequisites

- Basic Computer Knowledge
- C Programming Fundamentals (Preferred but Not Mandatory)

Module 1: Introduction to C++

Topics

- Evolution of Programming Languages
- Features of C++
- Difference Between C and C++
- Object-Oriented Programming Concepts
- Structure of C++ Program
- Compilers and IDEs
- Writing First C++ Program
- Lab
- Hello World Program
- Input and Output Program

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Module 2: Basics of C++

Topics

- Tokens
- Keywords
- Identifiers
- Variables
- Constants
- Data Types
- Operators
- Type Conversion
- Lab Programs
- Arithmetic Operations
- Area of Circle
- Temperature Conversion
- Interest Calculation

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Module 3: Input and Output Streams

Topics

- cin
 - cout
 - endl
 - Manipulators
 - Formatting Output
 - Lab Programs
 - Student Information Display
 - Employee Details
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Module 4: Control Statements

Topics

- if Statement
 - if-else Statement
 - Nested if
 - else-if Ladder
 - switch Statement
 - Lab Programs
 - Largest Number
 - Grade Calculation
 - Calculator Program
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Module 5: Looping Statements

Topics

- for Loop
- while Loop
- do-while Loop
- Nested Loops
- break
- continue
- Lab Programs
- Factorial
- Fibonacci Series
- Prime Numbers
- Armstrong Number



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- Multiplication Tables
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Module 6: Arrays

Topics

- One-Dimensional Arrays
 - Two-Dimensional Arrays
 - Array Operations
 - Lab Programs
 - Sum of Array Elements
 - Largest Element
 - Matrix Addition
 - Matrix Multiplication
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Module 7: Strings

Topics

- Character Arrays
 - String Class
 - String Functions
 - String Manipulation
 - Lab Programs
 - String Reverse
 - Palindrome Check
 - Count Vowels
 - String Comparison
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Module 8: Functions

Topics

- Function Declaration
- Function Definition
- Function Calling
- Function Overloading
- Inline Functions
- Default Arguments
- Recursion
- Lab Programs
- Factorial using Function



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- Fibonacci using Recursion
 - Function Overloading Examples
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Module 9: Object-Oriented Programming Fundamentals

Topics

- Classes and Objects
 - Data Members
 - Member Functions
 - Access Specifiers
 - Encapsulation
 - Lab Programs
 - Student Class
 - Employee Class
 - Product Class
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Module 10: Constructors and Destructors

Topics

- Default Constructor
 - Parameterized Constructor
 - Copy Constructor
 - Destructor
 - Lab Programs
 - Student Record System
 - Product Inventory System
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Module 11: Inheritance

Topics

- Single Inheritance
 - Multilevel Inheritance
 - Multiple Inheritance
 - Hierarchical Inheritance
 - Hybrid Inheritance
 - Lab Programs
 - Employee Payroll System
 - College Management System
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Module 12: Polymorphism

Topics

- Function Overloading
- Operator Overloading
- Function Overriding
- Virtual Functions
- Lab Programs
- Calculator using Overloading
- Shape Area Calculation

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Module 13: Friend Functions and Friend Classes

Topics

- Friend Function
- Friend Class
- Lab Programs
- Access Private Data Example

Module 14: Static Members

Topics

- Static Variables
- Static Functions
- Lab Programs
- Object Counter Program

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Module 15: Pointers

Topics

- Pointer Basics
- Pointers and Arrays
- Pointer to Objects
- this Pointer
- Lab Programs
- Swap Using Pointers
- Dynamic Object Creation

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Module 16: Dynamic Memory Management

Topics



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- new Operator
- delete Operator
- Dynamic Arrays
- Lab Programs
- Dynamic Student Records
- Dynamic Employee Database

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Module 17: File Handling

Topics

- File Streams
- ifstream
- ofstream
- fstream
- Reading and Writing Files
- Lab Programs
- Student Database
- Employee Information System

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Module 18: Exception Handling

Topics

- try
- catch
- throw
- Multiple Catch Blocks
- Lab Programs
- Division by Zero Exception
- Bank Account Validation

Module 19: Templates

Topics

- Function Templates
- Class Templates
- Lab Programs
- Generic Calculator
- Generic Array Operations

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Module 20: Standard Template Library (STL)

Topics

- Containers
 - Vector
 - List
 - Queue
 - Stack
 - Map
 - Set
 - Iterators
 - Algorithms
 - sort()
 - find()
 - count()
 - Lab Programs
 - Student Record Management using Vector
 - Library Management using Map
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Module 21: Advanced OOP Concepts

Topics

- Abstract Classes
 - Pure Virtual Functions
 - Interfaces
 - Runtime Polymorphism
 - Lab Programs
 - Banking System
 - Shape Hierarchy
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Module 22: Searching and Sorting

Topics

- Linear Search
- Binary Search
- Bubble Sort
- Selection Sort
- Insertion Sort
- Lab Programs



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- Student Search System
 - Marks Ranking System
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Module 23: Mini Projects

Project 1

Student Management System

Project 2

Library Management System

Project 3

Bank Management System

Project 4

Hospital Management System

Project 5

Employee Payroll Management System

Placement-Oriented Problem Solving

Topics

- Time Complexity Basics
 - Arrays Problems
 - String Problems
 - Searching Techniques
 - Sorting Techniques
 - Basic Competitive Programming
 - Practice Problems
 - 50 Basic Problems
 - 50 Intermediate Problems
 - 25 OOP-Based Problems
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Learning Outcomes

After completing this course, students will be able to:

- ✓ Develop Object-Oriented Applications
- ✓ Apply OOP Concepts in Real Projects
- ✓ Work with Classes, Objects, Inheritance, and Polymorphism
- ✓ Handle Files and Exceptions
- ✓ Use STL Containers and Algorithms
- ✓ Build Mini Projects Independently



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✓ Prepare for Campus Placements and Technical Interviews

Recommended Practice Plan

Activity	Count
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Basic Programs	75
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OOP Programs	50
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STL Programs	25
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Assignments	10
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Mini Projects	5
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Assessments	5
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Total Hands-on Programs: 150+

Mini Projects: 5

Course Level: Beginner to Advanced

Target Audience: B.Tech, Diploma, MCA, BCA, Job Seekers, Campus Placement Aspirants.